Batch mode: define a dataset, process it, read it back

Developer quick-sheet · goldsluice and goldpan share one dataset file · version 2, 6 November 2026

The full guide is now part of the goldsluice documentation: https://villano-lab.github.io/goldsluice/guides/batch.html. It has the whole definition file, every option, the messages and what to do about them. This sheet is the overview.

A **dataset definition** is one small text file that says *which raw files*, *how to read them* and *which processing to run*. `goldsluice` runs it over every raw file in one command and writes the results next to it; `goldpan` opens the same file and gives you the pulses together with the processed quantities. Nothing is assumed: the readout, the channels and every parameter are written in the file.

The loop

1. Write `<name>.dataset.toml`: the `[dataset]` name and output folder, one or more `[[sources]]` (format, files, detector, readout) and the `[[processing]]` steps in order.
2. Check and run it:

```
python -m goldsluice algorithms
python -m goldsluice check r76.dataset.toml
python -m goldsluice run r76.dataset.toml
```

3. Read it back in a notebook:

```
from goldpan.dataset import Dataset

ds = Dataset.open("r76.dataset.toml")
ev = ds.read()          # the pulses with the processed quantities
q = ds.processed()      # only the quantities: small and fast
```

What to remember

- **Results do not belong in the repository.** Write them outside it (`~/goldsluice-output/`, or set `GOLDSLUICE_OUTPUT`); `check` and `run` refuse a folder that Git would track.
- **Unknown keys are errors**, so a typo fails at once.
- A raw file is processed again only when it, its reading, its readout or its steps change.
- Steps can build on each other (a filter, then `peak` on its result), several sources can share one file, run metadata (high voltage, trigger setting) is joined when the data are read, and the quantities can be written as ROOT files.

Where things are

You want	Go to
the guide	https://villano-lab.github.io/goldsluice/guides/batch.html
the file format, as a specification	the docstring of `goldpan.dataset`, https://villano-lab.github.io/goldpan/reference/generated/goldpan.dataset.html
example definitions	`examples/` in https://github.com/villano-lab/goldsluice
a new algorithm to name in a step	the sheet *Running goldsluice and developing an algorithm*